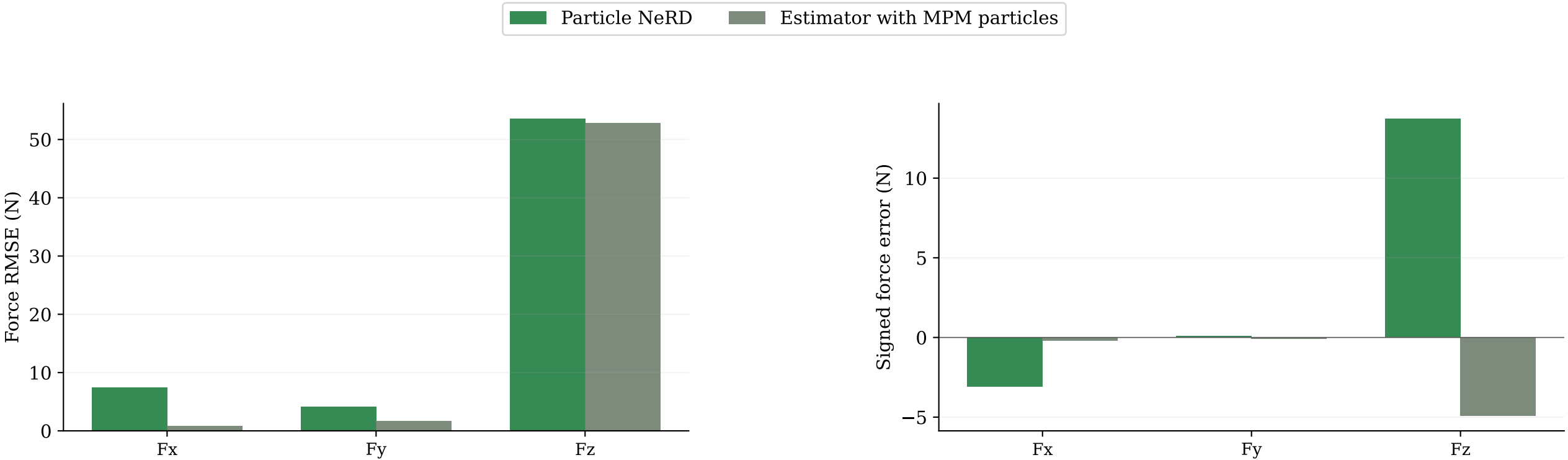

Figure 8: Frozen object-model checkpoint 9,000. Box: fixed-object hold; 23 s recording; one recorded-motion replay. Left: component RMSE; right: signed mean error (estimate minus native MPM material force) over the full recording. World-frame forces, +z upward. Raw 200 Hz forces, no filtering or time shifts. The gray diagnostic uses the same estimator with MPM particles, not a second neural model.